

UNIT-3

OPTICAL
TRANSMITTERS

Richa
Menta
AP (ECE)

INTRODUCTION

①

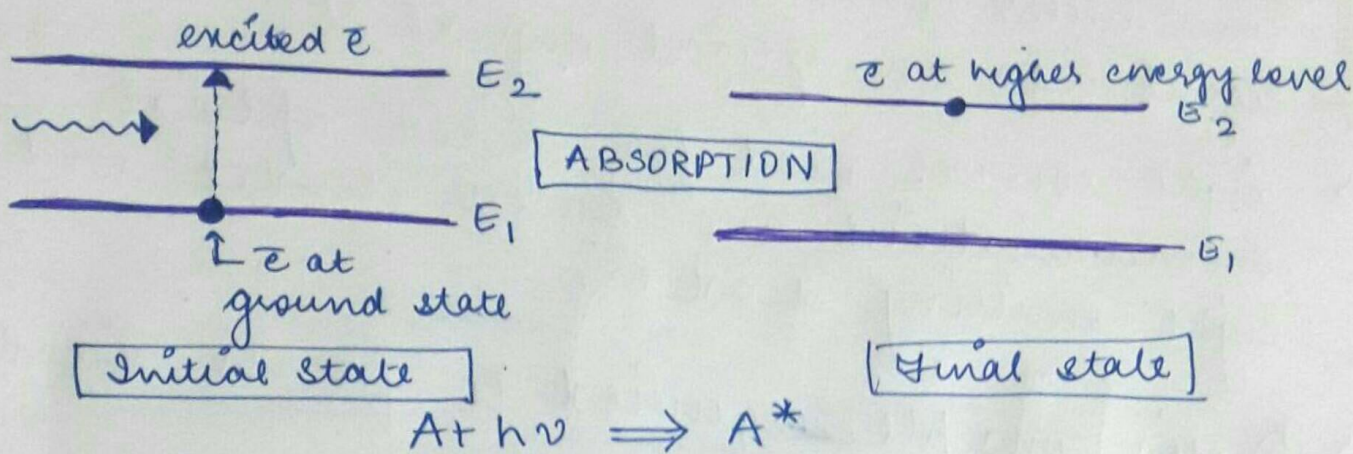
- In optical communication system, optical transmitter converts the incoming electrical signal into an optical signal.
- Light sources used for fiber optical communications are laser diodes and light emitting diodes.
- Basic concepts required to form laser Beam

- ABSORPTION
- SPONTANEOUS EMISSION
- STIMULATED EMISSION

① Absorption

- Under normal condition, the tendency of electrons is to stay at the lower energy level.
- This lower energy level is called as ground state (E_1).
- If a photon of certain energy is incident on the electron, then the e^- absorbs this energy.

→ So, the total energy contained by that $\bar{e}$ gets increased and now it is possible for that $\bar{e}$ to shift towards higher energy level (E_2).



→ Here, the important consideration is that, the incident ~~light~~ photon should provide energy atleast $E_2 - E_1$, for an $\bar{e}$ to make transition from ground state to higher level state.

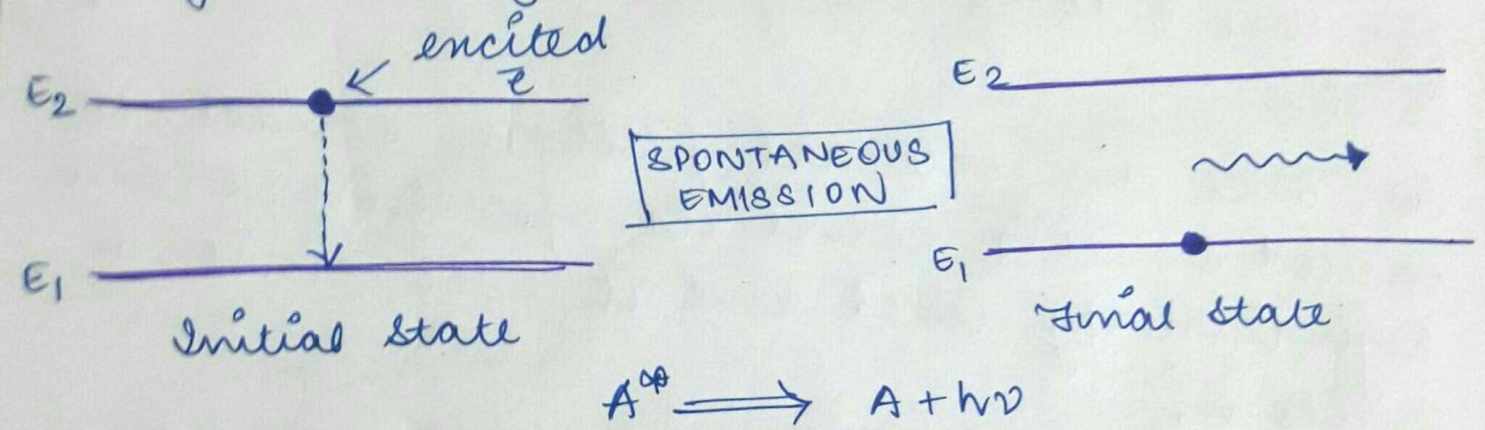
→ This process is referred to as Stimulated Absorption.

② Spontaneous Emission

→ After the absorption process, the electrons present at the higher energy level are in excited state.

→ Now, if no photons are incident externally, then these $\bar{e}$ s stay at this stage for a short time.

- But after some time period, the e^- falls back to its original state i.e. at lower energy level.
- consider one e^- present at the higher energy level. when this e^- falls back to the lower energy level, then it will emit the energy which was absorbed by it during the absorption process.



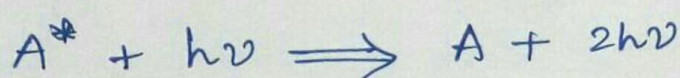
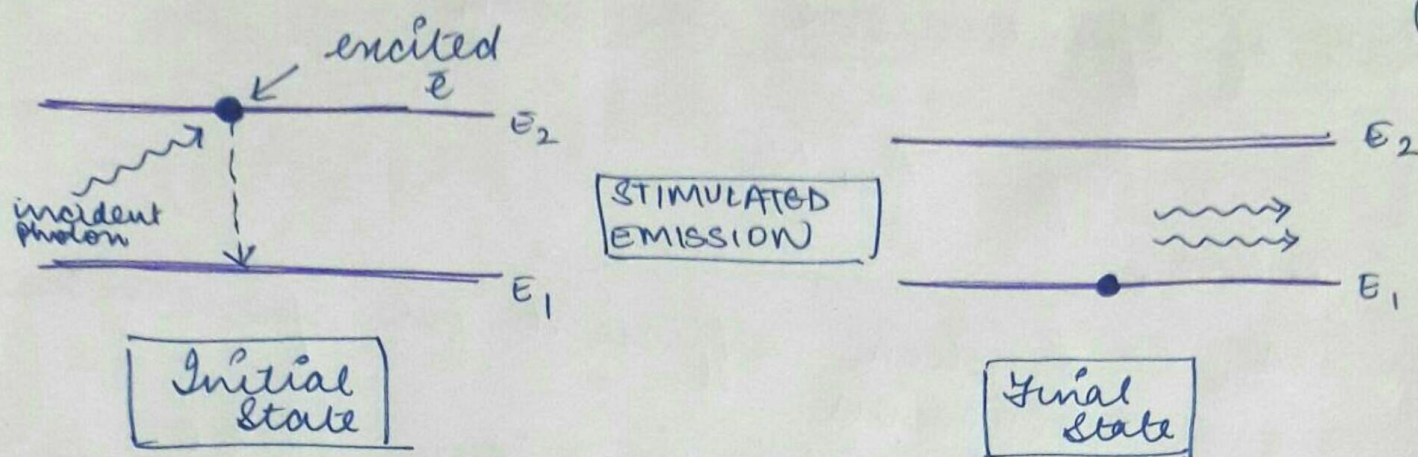
- The energy emitted during this transition of electron from E_2 to E_1 is simply the energy level difference between these two energy levels.
- This process is referred to as Spontaneous Emission.

③ Stimulated Emission

④

- This is the key process of the formation of laser beam.
- In this process, amplification of the light takes place.
- If an incident photon is allowed to strike an excited e^- , then there is no chance for this e^- to go to the further higher energy level.
- So, once an external photon strikes to excited e^- , this leaves its position from higher energy level and it will emit a photon.
- Thus, we are getting two photons at the output, one which is striking on an excited e^- , and other coming out of excited e^- leaving the higher energy level.
- So, one photon causes emission of two photons and the light amplification takes place.
- These two photons are in the same phase and travelling in the same direction.

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→ The process in which the number of excited es or atoms are made larger than that of es or atoms present at ground state.

This is known as POPULATION INVERSION

ABSORPTION & EMISSION RATES

Rate of Absorption

$$R_{abs} = B_{12} \rho(\nu) N_1$$

where,

$N_1 \rightarrow$ population at lower level
 $\rho(\nu) \rightarrow$ energy density of incident light
 $B_{12} \rightarrow$ constant of proportionality known as Einstein coefficient of Absorption

Rate of Spontaneous Emission

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$$R_{sp} = A_{21} N_2$$

where,

$A_{21} \rightarrow$ Einstein coefficient for spontaneous emission

$N_2 \rightarrow$ population at higher level

Rate of Stimulated Emission

$$R_{st} = B_{21} \rho(\nu) N_2$$

where,

$N_2 \rightarrow$ population at higher level

$\rho(\nu) \rightarrow$ energy density of incident light

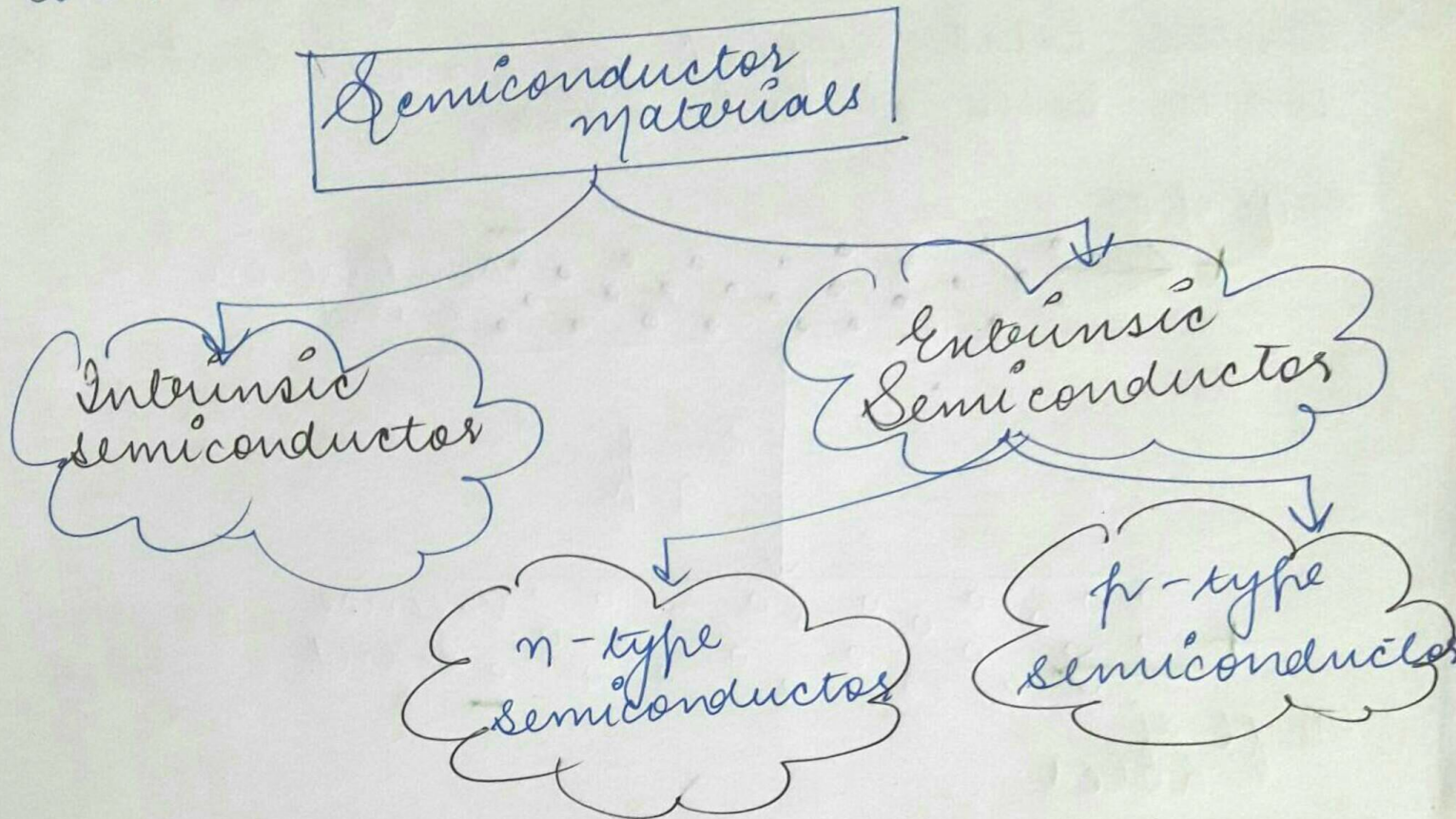
$B_{21} \rightarrow$ Einstein coefficient for stimulated emission

NOTE

- $\rightarrow$ Einstein found that the emission of a photon is possible by 2 different processes:-
- Spontaneous & stimulated emission.
- and the coeff's describing these processes:-
- Absorption, spont. and stimulated emission
- $\rightarrow$ Einstein coeff's are a measure of probability of Absorption or Emission of light by an e or atom.

SEMICONDUCTOR MATERIALS (7)

A semiconductor material is one whose conductivity lies between that of a conductor and an insulator.



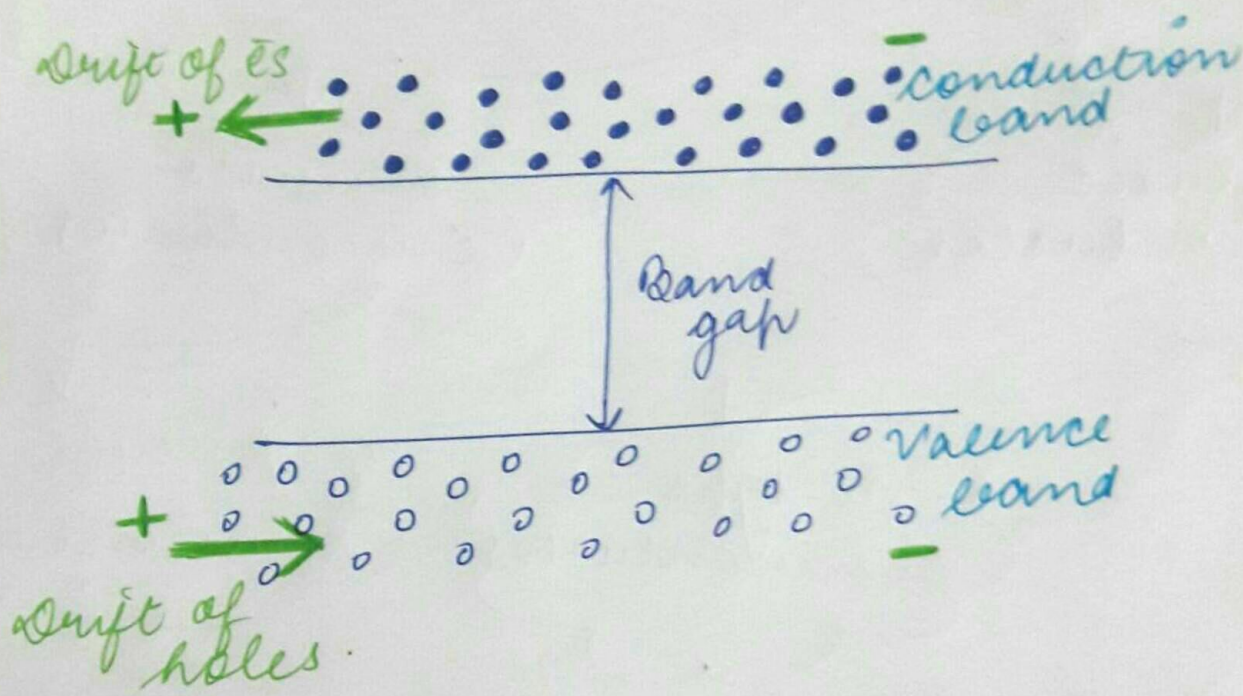
① Intrinsic Semiconductor

- A semiconductor in an extremely pure form is known as intrinsic semiconductor. (eg:- Si, Ge).
- Here, no. of electrons are equal to the no. of holes.
- when an external electric field is applied across an intrinsic semiconductor, the conduction occurs due to both

holes and electrons.

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- The free e^- s in conduction band move towards the +ve terminal of the battery and holes in valence band move towards the -ve terminal of the battery.
- So, the total current in semiconductor is due to both e^- s and holes.



Intrinsic Semiconductor

n-type
Semiconductors

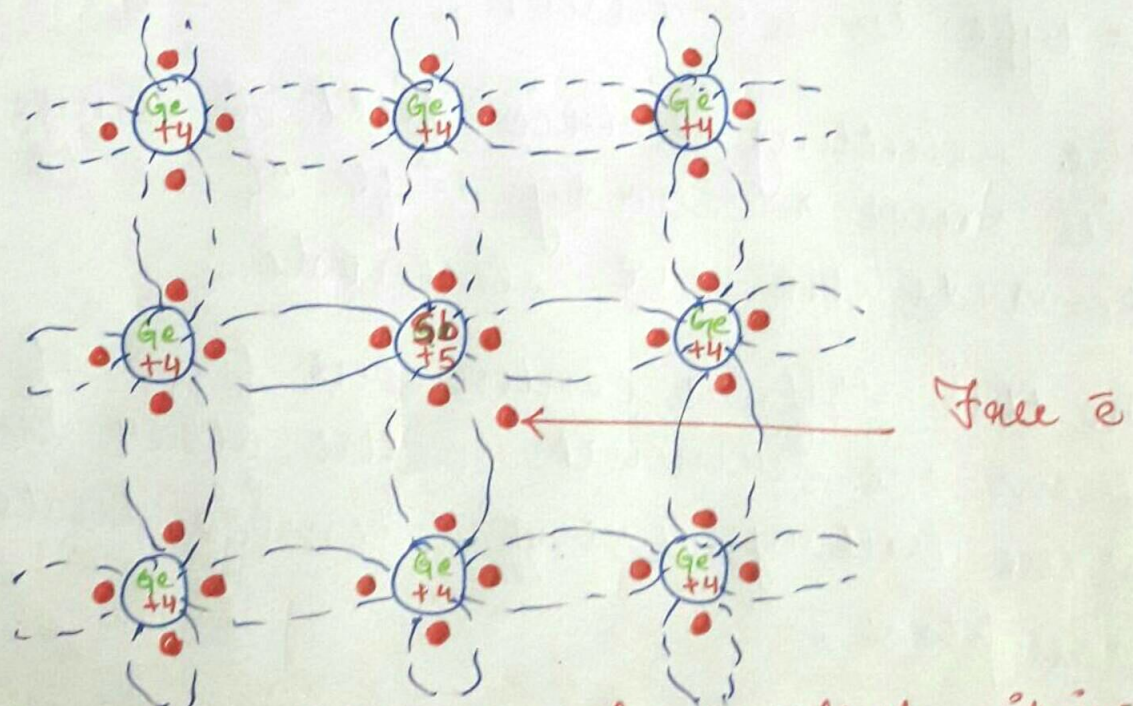
p-type
Semiconductors

→ The type of semiconductor in which impurity is added is known as **Intrinsic semiconductor** or **Impure Semiconductor**.

- The process of adding the impurity is (9) known as doping.
- The impurity which is added is known as dopant.

a n-type Semiconductor

- * When a pentavalent impurity is added to a pure semiconductor, then it is known as n-type semiconductor.
- * Examples of pentavalent impurities are Phosphorus (P), Antimony (Sb), Arsenic (As).
- * Pentavalent impurities have five valence e^- s, out of which four e^- s will make the covalent bonds with the four valence e^- s of neighboring host crystal atoms.

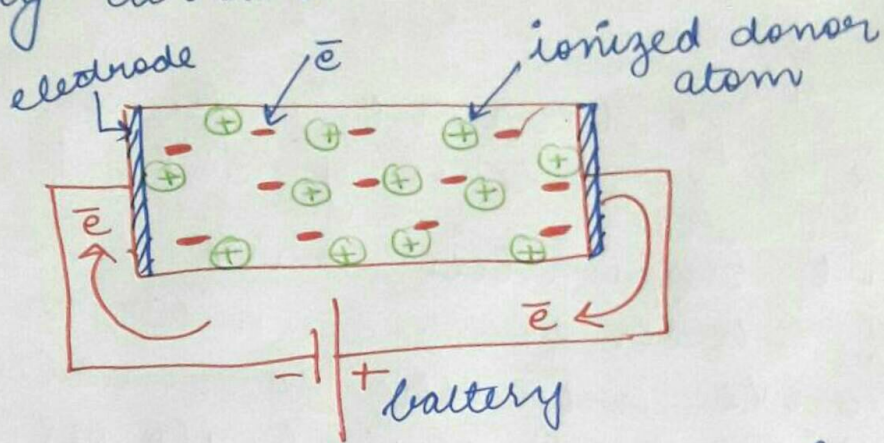


Crystal lattice of (Ge) when pentavalent impurity (Sb) added.

* Like this, there will no. of excess free e^- (10) which are known as donors when n-type impurities has been added in Ge.

* The e^- being negatively charged particles.
 these semiconductors are known as n-type semiconductors.

* In n-type semiconductors, e^- are the majority carriers and the holes are the minority carriers.



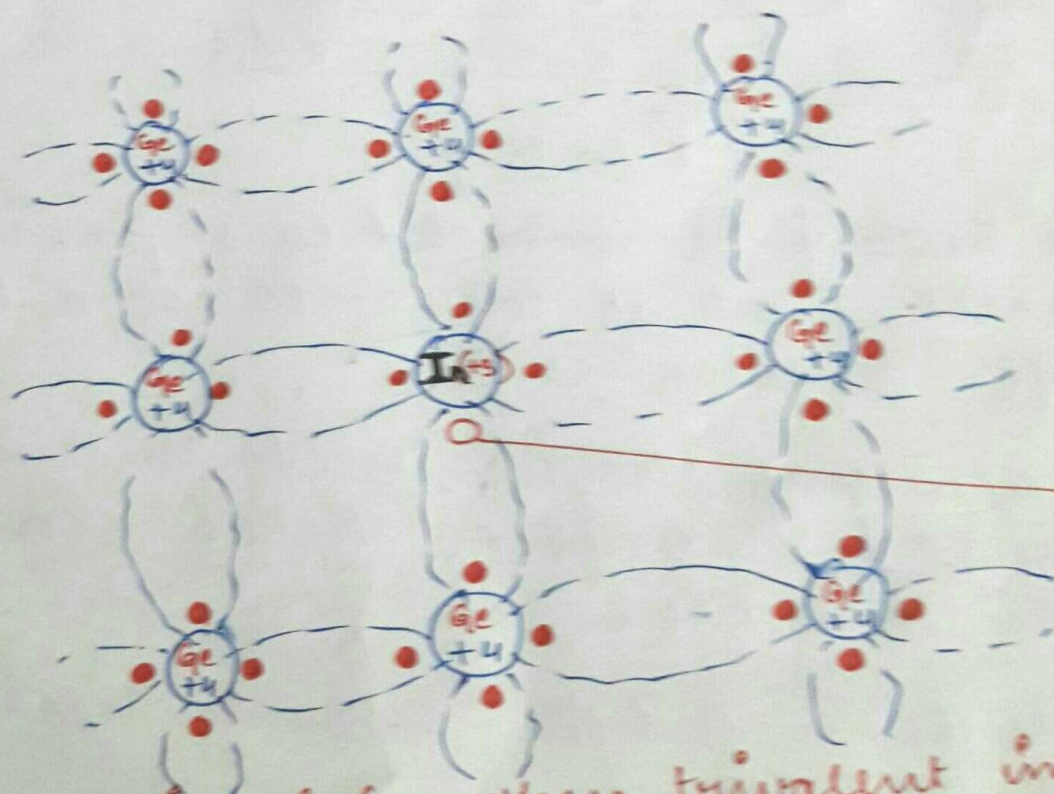
* Let a battery be connected to a pair of electrodes fitted at the ends of a bar of n-type semiconductor.

* The resulting electric field in semiconductor will cause a steady drift of e^- to move towards the +ve electrode.

* In this way, a continuous flow of e^- from one terminal to the other terminal of the battery through semiconductor bar takes place.

p-type Semiconductor

- * When a trivalent impurity is added to a pure semiconductor, then it is known as p-type semiconductor.
- * Examples of trivalent impurities are Indium (I), Boron (B), all aluminium (Al), Gallium (Ga).
- * Trivalent impurities have three valence e^- , so all the three e^- makes covalent bonds with three valence e^- of neighbouring host crystal atoms.
- * But, one bond is incomplete due to less electron.

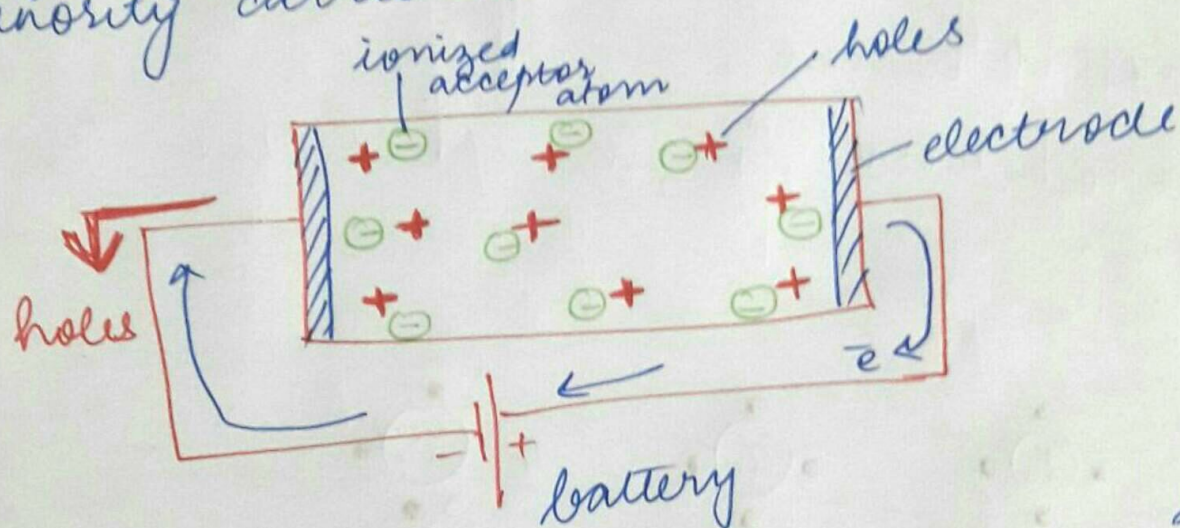


Crystal lattice of Ge when trivalent impurity (In) is added

* Like this, there will be large no. of holes w/c are known as acceptor ⁽¹²⁾ when p-type impurity has been added in Ge.

* The holes being positively charged $\therefore$ these semiconductors are known as p-type semiconductors.

* In p-type semiconductors, holes are the majority carriers and the e^- are the minority carriers.



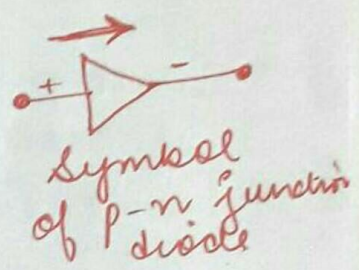
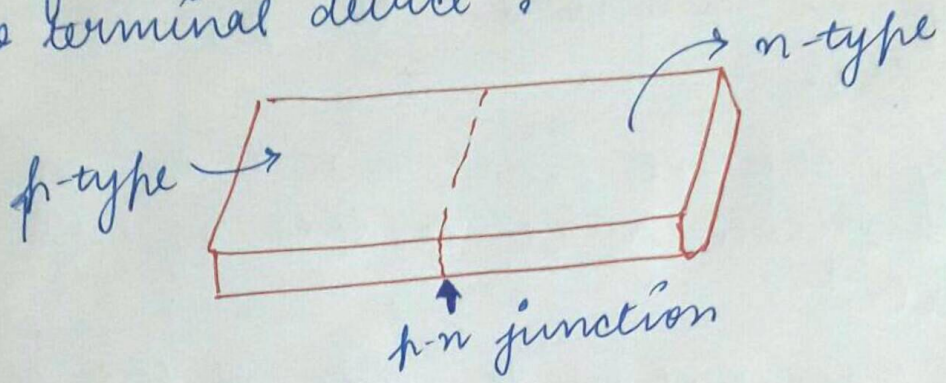
* Let a battery be connected to a pair of electrodes fitted at the ends of a bar of p-type semiconductor.

* The resulting electric field in semiconductor will cause a drift of holes towards the -ve electrode.

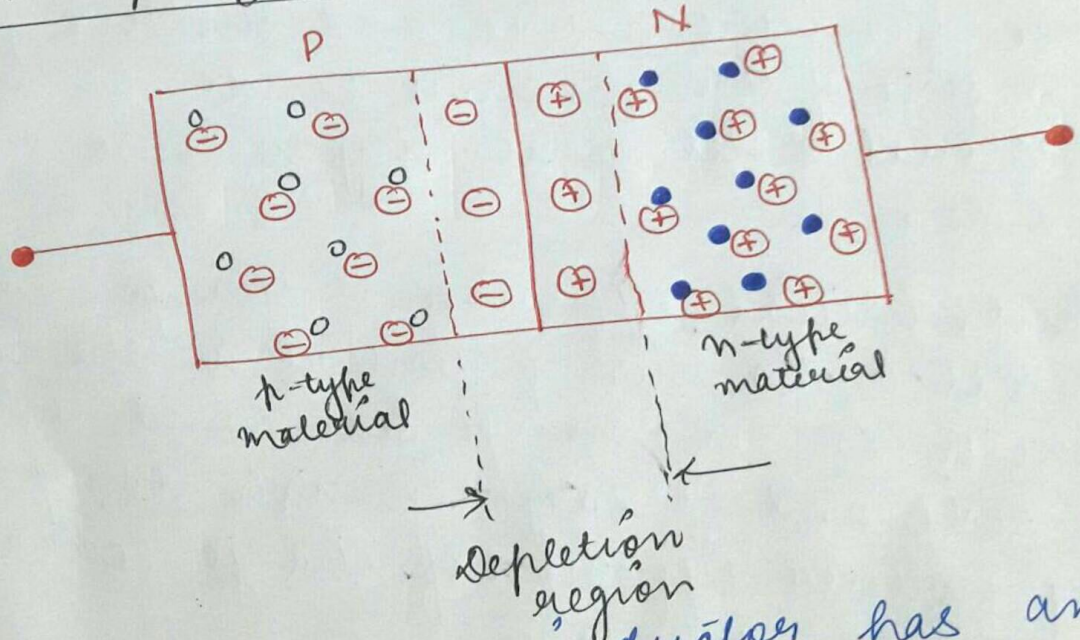
* $\therefore$ current in p-type semiconductor is carried by holes.

p-n junctions

- * A junction between a p-type and a n-type semiconductor is known as pn junction.
- * When one side of a single crystal semiconductor is doped with acceptors and the other side with donors, a p-n junction is produced.
- * It is referred to as a junction diode and is a two terminal device.



① Unbiased pn junction



→ The p-side of the semiconductor has an excess of holes and n-side of the semiconductor has an excess of electrons.

→ When a p-n junction is formed, some (14) free e^- from n-side diffuse across the junction to combine with holes on p-side.

→ The free e^- crossing the junction creates $-ve$ ions on p-side and also leaves behind positive ions on the n-side.

→ Now, as the $-ve$ ions are created on the p-side of the junction, the p-side acquires $-ve$ potential.

→ Similarly, the $+ve$ ions created on n-side of the junction gives n-side a $+ve$ potential.

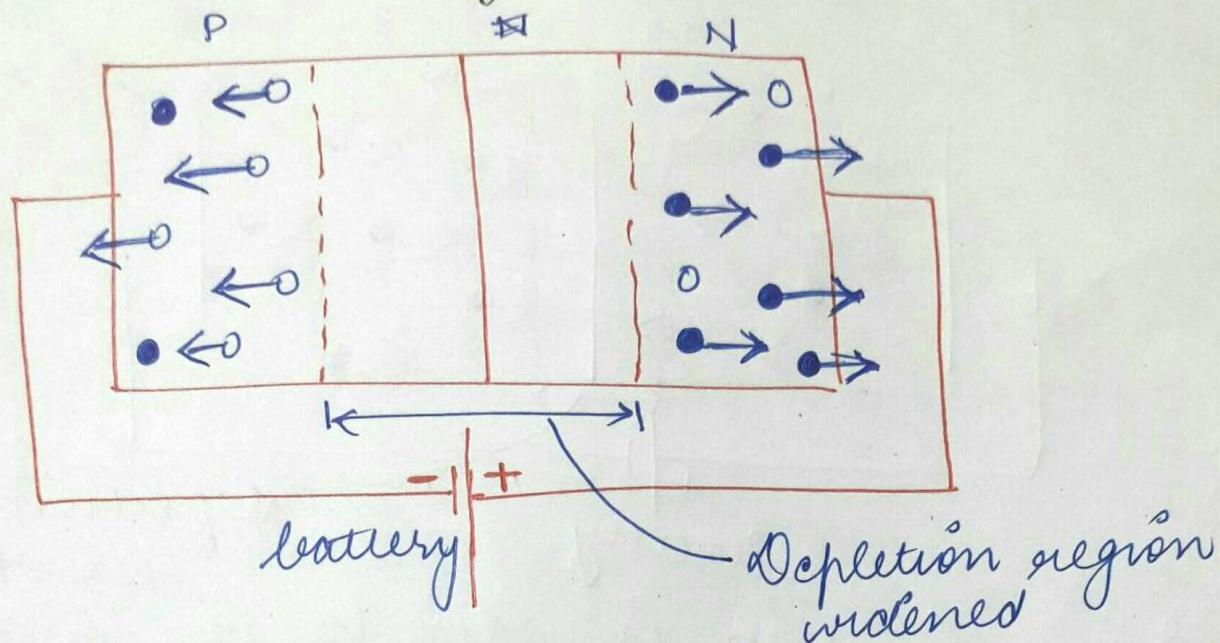
→ The $-ve$ potential on p-side of the junction tends to repel additional e^- that try to come from the n-side and the $+ve$ potential on n-side of the junction tends to repel additional holes that try to come from p-side.

→ Thus, initial diffusion of charge carriers creates a barrier potential at the junction which is $-ve$ on p-side and $+ve$ on n-side and is large enough to prevent any further diffusion of charge carriers across the junction.

→ The movement of charge carriers across the junction leaves a layer on each side which is depleted of charge carriers. Known as ~~barrier~~ Depletion Region.

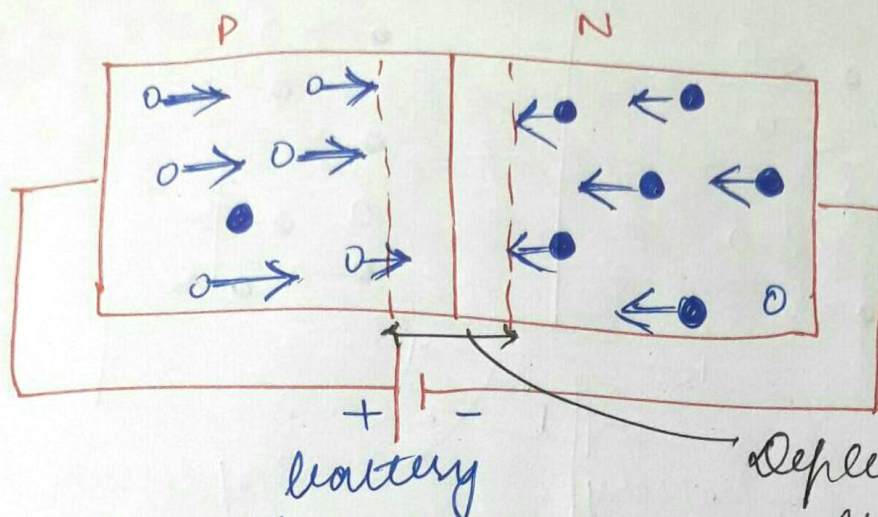
2) Reverse biased P-N junction

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- A p-n junction is said to be reverse biased when p-side is connected to -ve terminal of the battery and n-side is connected to +ve terminal of the battery.
- e^- and holes from n-side and holes from p-side are attracted away from the junction.
- This will result in widening of depletion region.
- There is no possibility of current flowing through the junction due to majority carriers but a small current can flow due to the minority carriers on each side.
- Therefore, a small amount of current i.e. reverse saturation current flows.

③ Forward biased junction



- A p-n junction is said to be forward biased when p-side is connected to +ve terminal of the battery and n-type is connected to -ve terminal of the battery.
- +vely charged holes are attracted towards the -ve terminal of the battery and -vely charged e^- s are attracted towards the +ve terminal of the battery.
- As movement of the charged carriers is towards the junction and the result is that depletion region is reduced and the barrier potential is also reduced.
- Due to reduction in barrier potential more and more majority charge carriers can diffuse across the junction which constitutes continuous flow of current.
- This current is known as forward current.

Non-Radiative Recombinations

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- The excess carriers can recombine either radiatively or non-radiatively.
- In radiative recombination, a photon is emitted and the non-radiative recombination effect has optical absorption in the active region.
- The non-radiative recombination effect is given by internal quantum efficiency (η).
- Internal quantum efficiency in the active region is the ratio of radiative recombination rate to the total recombination rate,

→ If

$R_r \rightarrow$ radiative recombination rate

$R_{nr} \rightarrow$ non-radiative recombination rate.

$$\therefore \eta_i = \frac{R_r}{R_r + R_{nr}}$$

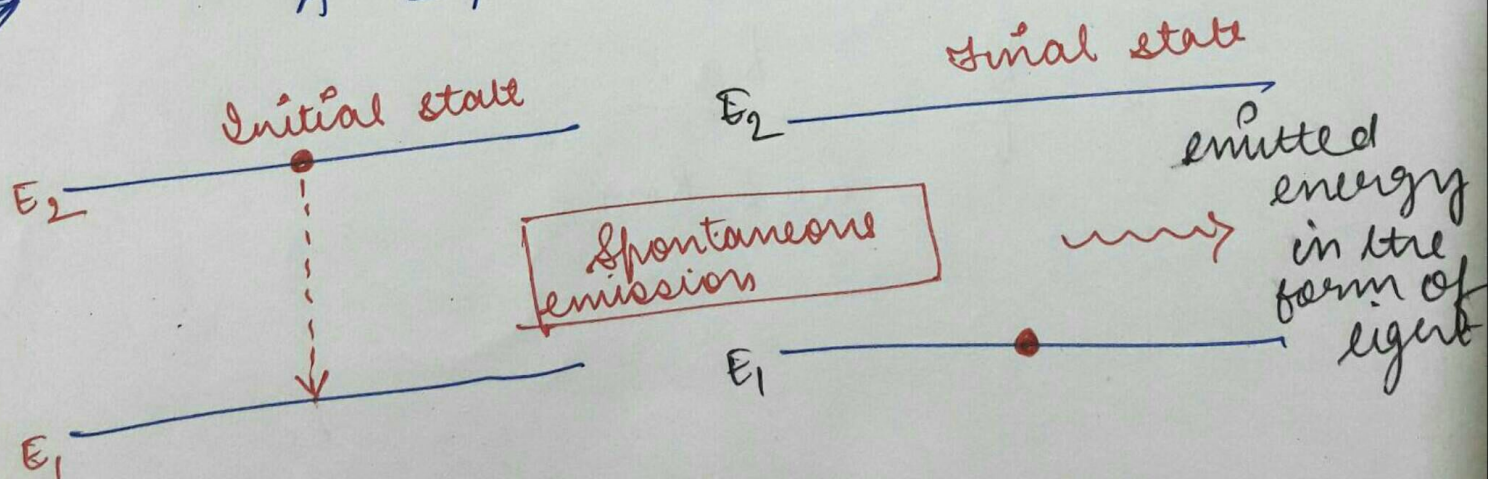
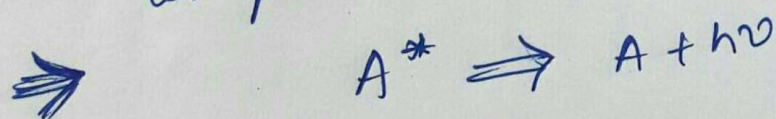
LED (Light Emitting Diode)

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- * LED is a semiconductor light source that emits light when current flows through it.
- * The major uses of LED are to illuminate objects and even places.
- * Its application is everywhere due to its compact size, low consumption of energy, extended lifetime, flexibility in terms of use in various applications.

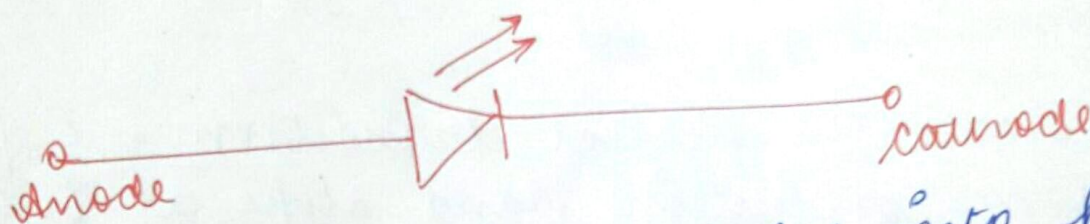
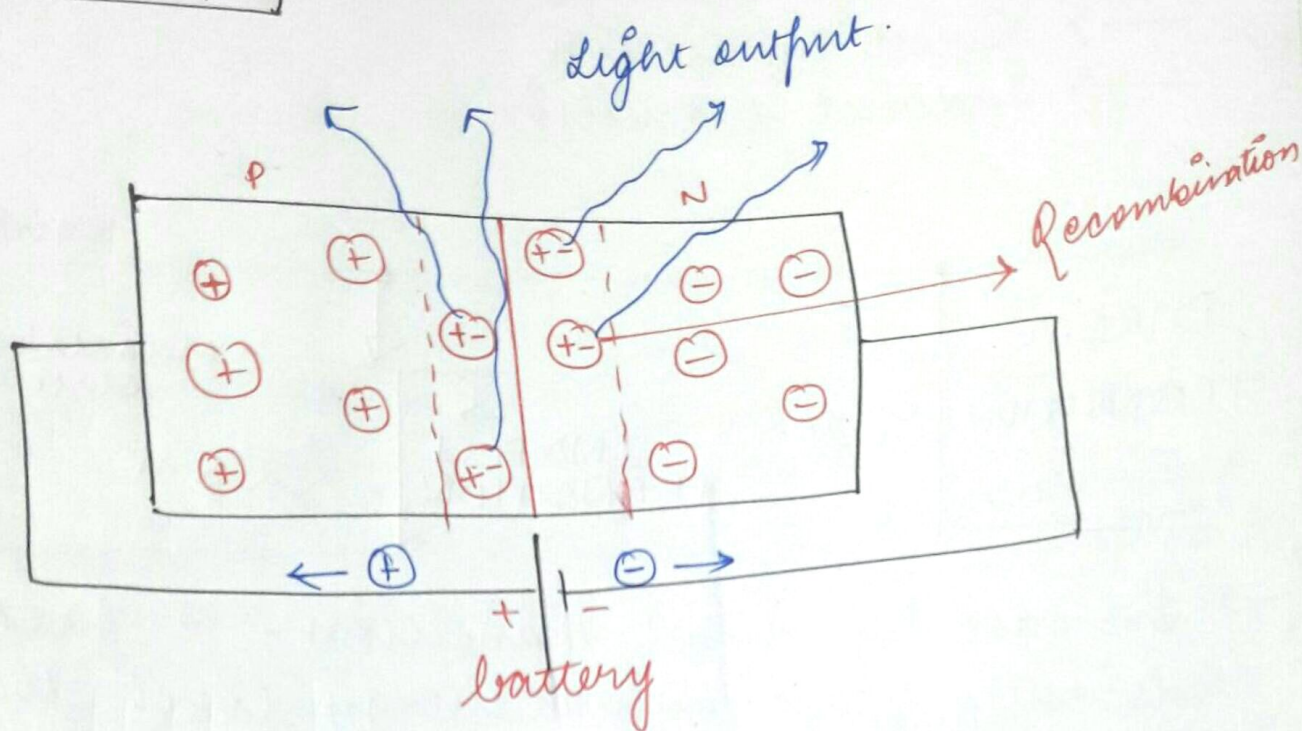
PRINCIPLE

- It is based on the principle of spontaneous emission.
- When a free electron from the higher energy level gets recombined with the hole, then it gives out the light as output.



WORKING

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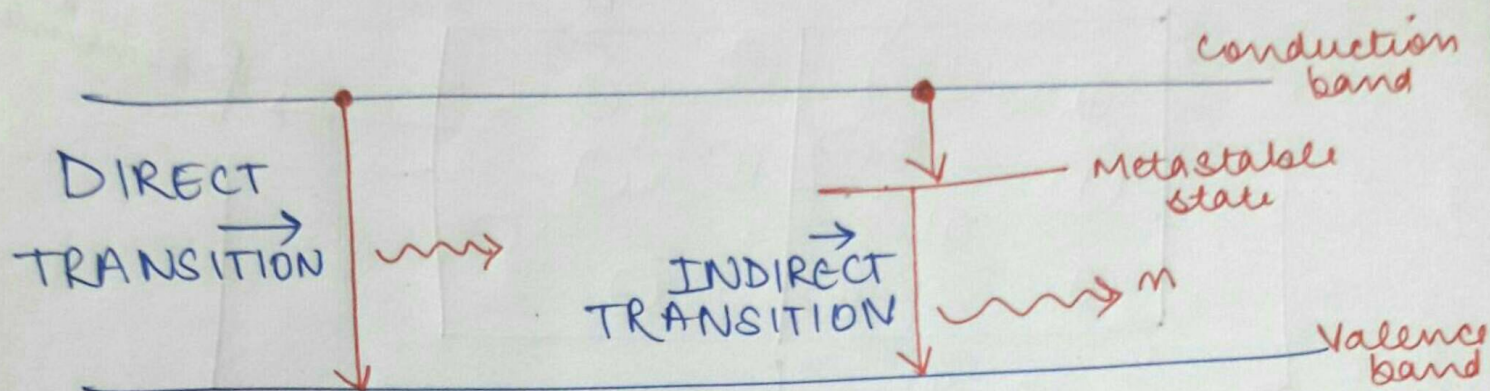


- The battery supplies the holes into p-region and e^- into n-region.
- The holes and e^- supplied by the battery are known as injected charge carriers.
- near the barrier potential, injected e^- combines with holes.
- This means that the e^- falls from higher energy level to lower energy level.
- when this transition occurs, the energy is radiated out.

Generally, two types of transition takes place (20)

→ Direct transition

→ Indirect transition

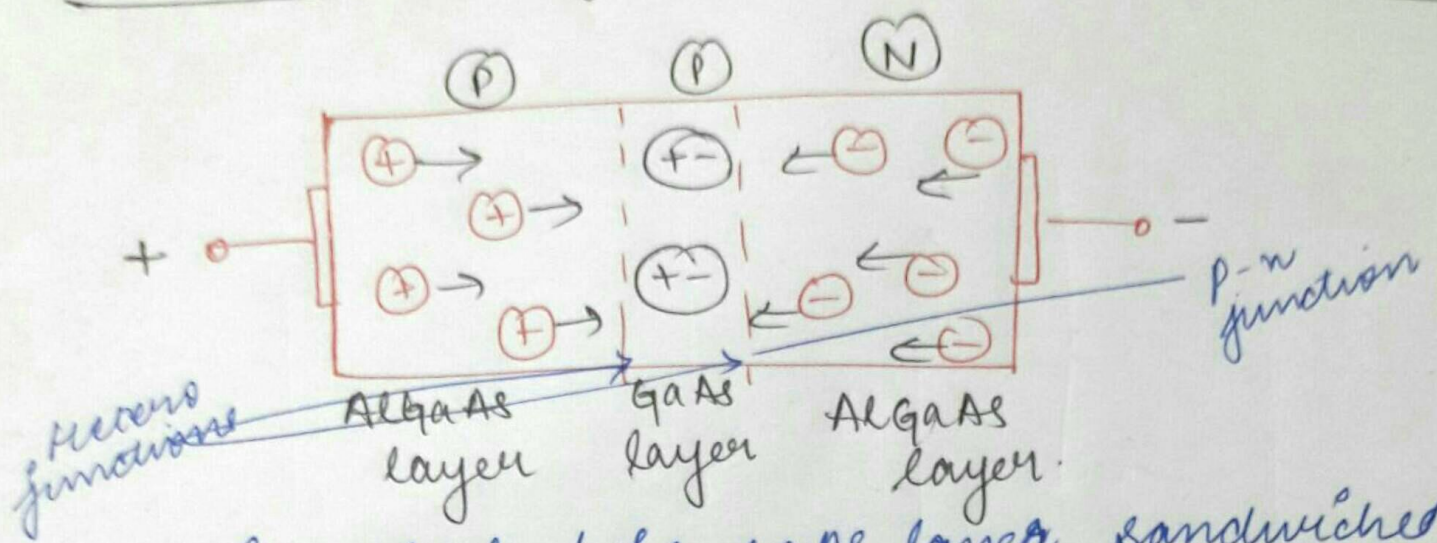


→ In case of direct transition, e^- falls directly from higher energy level to the lower energy level.

→ In case of indirect transition, e^- falling from conduction band stops at intermediate energy level (called as metastable state) for a certain time and then it will reach to valence band.

→ So, definitely time reqd. for indirect transition is more than that of direct transition.

Double Heterojunction LED (DHLED) (24)



- It consists of p-type GaAs layer sandwiched between p-type AlGaAs layer and n-type AlGaAs layer.
- The LED is forward biased by connecting p-type AlGaAs layer to the +ve terminal of the battery and n-type AlGaAs is connected to the -ve terminal of the battery.
- When LED is forward biased, e^- are injected from n-type ~~AlGaAs~~ to p-type GaAs.
- After crossing the p-n junction, these e^- become the minority carriers.
- The e^- then recombine with holes and the light is emitted out.

Advantages

- Good quantum efficiency
- High Radiation emission

Disadvantage

Such LEDs have low coupling efficiency to optical fibres.